Bachelor Thesis Proposal

Automated Scene-Level Feature Annotation Tool for EGY-DRiVeS Dataset

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1 Problem Formulation

For autonomous vehicles to work efficiently, they require high-quality labeled datasets to improve their perception and decision-making process. Manually annotating the scenes is very time-consuming and error-prone because the EGY-DRiVeS dataset provides multi-sensor data that requires high accuracy. This thesis aims to develop an automated tool that extracts and annotates scene-level features efficiently, ensuring consistency and accuracy in the dataset.

2 Thesis Objectives

The main objectives of this thesis are:

- Develop an automated annotation tool for scene-level features in the EGY-DRiVeS dataset.
- Implement feature extraction techniques for traffic conditions, road types, and environmental contexts.
- Ensure high annotation accuracy and consistency compared to manual labeling.

3 Thesis Propositions

This research is based on the following key propositions:

- Automated feature annotation can significantly reduce the time and effort required for dataset labeling.
- Machine learning and image processing techniques can accurately identify and classify scene-level features.
- High-quality automated annotations will enhance autonomous vehicle perception models trained on the EGY-DRiVeS dataset.